1. **What is a Class in Python, and How is it Used to Create Objects?**

Ans:

A class is a blueprint or template used to create objects. It defines the properties (attributes) and behaviors (methods) that objects of that class will have.

An object is an instance of a class.

Syntax

```
class ClassName:
    pass
```

Example

```
class Student:
    def __init__(self, name):
        self.name = name

student1 = Student("Rahul")
print(student1.name)
```

Output:

```
Rahul
```

In the above example:

`Student` is a class.

`student1` is an object of the class.

`"Rahul"` is assigned to the object's attribute.

Advantages of Classes

Code reusability.

Better organization of data and functions.

Supports Object-Oriented Programming (OOP).

2. **What are Methods and Attributes in Python Classes?**

Ans:

Attributes

Attributes are variables that store data related to an object or class.

Example

```
class Student:
    def __init__(self, name, age):
        self.name = name
        self.age = age
```

Here:

`name` and `age` are attributes.

Methods

Methods are functions defined inside a class that describe the behavior of objects.

Example

```
class Student:
    def __init__(self, name):
        self.name = name

    def display(self):
        print("Student Name:", self.name)

student1 = Student("Rahul")
student1.display()
```

Output:

```
Student Name: Rahul
```

Here:

`display()` is a method.

It performs an action using the object's data.

Difference Between Attributes and Methods:

Attributes	Methods
Store data or properties of an object.	Define actions or behavior of an object.
Represent characteristics.	Represent functionality.
Example: `name`, `age`	Example: `display()`, `calculate()`

Example

```
class Car:
    def __init__(self, brand):
        self.brand = brand    # Attribute
    def start(self):         # Method
        print(self.brand, "is starting")
car1 = Car("Toyota")
car1.start()
```

Output:

```
Toyota is starting
```